DATA

Data Sufficiency, Data caselets & other imp. concepts

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S. No.	Topic	Pg. No.
1.	**Data Sufficiency**	
2.	1.1 Data Sufficiency	1
3.	1.2 Steps to solve Data Sufficiency Questions	1
4.	1.3 Data Sufficiency Tricks and Examples	3
5.	1.4 Data Sufficiency Examples	4
6.	**Data Caselets**	
7.	2.1 Data Caselets	13
8.	2.2 Data Caselets Examples	15
9.	**Data Comparison**	19
10.	3.1 Data Comparison Strategy	19
11.	3.2 Solved Examples	21

Data Sufficiency

1.1 Data Sufficiency

Data Sufficiency as the name signifies are the questions, where in you need to check whether the data given in the statements is sufficient or not to answer the question. In data sufficiency problems usually a question has two statements labeled A and B, which contain certain data. In these questions, the important point is that you need not to solve the question but to check whether the data given in the statements are sufficient to solve the question or not. The statement will be considered sufficient if the answer you obtain is unique. If a question cannot be solved with the help of the statement or it has multiple answers, then the given statement will be insufficient.

The first step for solving these questions is that you should take both the statements as the individual statement i.e. there is no relation in the statements. To solve the question, take the first statement and check whether the question can be solved by using this statement or not. If you get answer from first statement then it is sufficient.

Now come to the second step. If with the help of the information given in the second statement you are able to solve the question then this statement is also valid and mark your answer accordingly. There is a possibility that first statement has the required information to solve the question but not the second and vice versa. In this case you will mark the answer as only one statement is sufficient.

There is still a chance that you are not able to solve the question with the help of given statements individually. In this case, try to solve the question by combining the information given in both the statements together. If you get the answer, then mark your answer as both statements are required. If you do not get the answer, then mark your answer as data is insufficient. Keep one thing in mind that you cannot combine the information given in the statements as long as at least one of the statements is sufficient. Now let us discuss the format of Data Sufficiency questions in detail.

Each data sufficiency problem consists of a question and two statements labeled A and B, in which certain data are given. You have to decide whether the data given in the statements are sufficient for answering the question. Using the data given in the statements plus your knowledge of mathematics and everyday facts (such as the number of days in June or the meaning of counterclockwise), you mark the answer as

- if Statement A **alone** is sufficient, but statement B alone is not sufficient to answer the question asked;
- Statement B **alone** is sufficient, but statement A alone is not sufficient to answer the question asked;
- **if Both** statements A and B **together** are sufficient to answer the question asked; but **neither** statement **alone** is sufficient;
- **if Each** statement **alone** is sufficient to answer the question asked;
- if Statements A and B **together** are **not** sufficient to answer the question asked, and additional data sufficient to the problem are needed.

Solve the problem and indicate the best of the answer choices given.

1.2 Steps to solve Data Sufficiency Questions

Step 1 – Examine the Question

Before looking at the two numbered statements, take twenty to thirty seconds to consider the question by itself. Figure out what is being asked. There are generally two possibilities a specific number may be sought (“What is the value of x?” "How many gallons of water are in the tank?”), or a true/false answer may be needed (“Is it true that $a > 7$?” “Is n a prime number?”) Make sure you understand exactly what the question is asking.
Then consider what information would be needed to answer the question. This will depend on the type of question, of course. If it is a geometry question, the information needed will be based on rules you’ve learned about how one geometric fact can be deduced from another. For example, to determine the area of a circle, you need to know its radius, its diameter, or its circumference. To determine the length of the hypotenuse of a right triangle, you need to know the length of the other two sides.

On the other hand, if it is a percentage question, different rules will come into play. To determine what percentage *X* is of *Y*, for example, you need to know the value of *X* and the value of *Y*. When a change from one value to another is involved – the increase in value of an investment, for example – you need to know both the old value and the percentage by which it has increased if you want to calculate the new value.
As these examples suggest, the data sufficiency question format allows the test makers to measure your knowledge of a wide array of mathematical topics.

Step 2 – Consider each statement individually

Having figured out the nature of the question and decided, in a general way, what information is needed to answer it, look at each of the two numbered statements provided. Consider them one at a time, without reference to each other.
First look at statement A. Does it provide, all by itself, enough information to answer the question? If so, you've already narrowed the possible answer choices to just two: 1 and 4. If not, three answer choices are possible: 2, 3 and 5.
Then look at statement B. Does it provide, all by itself, enough information to answer the question? If so, only answers 2 and 4 are possible. If not, only answers 1, 3 and 5 are possible.
Having gotten this far, you may already be able to pick the right answer. If either statement by itself provides enough information to answer the question, you can pick from answers 1, 2 and 4, depending on which statement is sufficient or whether either statement will do.
If neither statement by itself is sufficient to answer the question, go on to the third stage:

Step 3 – Combine the two statements

Third, if necessary, combine the two statements. If neither statement by itself is sufficient to answer the question, consider whether you can answer the question by combining the information in both statements. If so, the answer is 3; if not, the answer is 5.

The following flow chart summarizes the questions you need to ask yourself as you use the three-stage system. It's a handy way to review and refresh your understanding of this method.

1.3 Data Sufficiency Tricks and Examples

Numbers: All numbers used are real numbers.
Figures: A figure accompanying a problem-solving question is intended to provide information useful in solving the problem. Figures are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. Straight lines may sometimes appear jagged. All figures lie in a plane unless otherwise indicated.

Figures are not necessarily drawn to scale

For data sufficiency questions, the directions state that the figures will confirm to the information given in the question stem. However, they will not necessarily confirm to the additional information provided in the numbered statements. So you cannot use a figure alone to estimate an answer, Note that this is contrary to the directions given for problem-solving sections, where figures are drawn to scale unless otherwise noted.

Do as little work as possible

The task is to make a judgment about the sufficiency of the additional information. Once you know that the information is sufficient to answer the question, the game is over. You don't need to go further and work out an actual numerical solution.

Unknowns can stand for any number

Unless otherwise indicated, an unknown ranges over the entire number line. It can be 0, 1, a fraction, any other positive number or any negative number.

Plugging in Numbers: negative, fraction, 0, 1

Many data sufficiency items include one or more unknown quantities, *x*, *n*, *a* and so forth. Just as with multiple – choice problems, there may be times when the easiest and quickest way to tackle such an item is by plugging a possible value into the equation or statement given and then working backward.
When you do this, don't forget to take into account the kind of "odd ball" values often overlooked: negative numbers, fractions, zero, and 1. Unless these are explicitly ruled out, they are possible values for the unknown, and the variations they introduce can turn an "answerable" question into an "unanswerable" one.

Don't Assume

One of the logical skills that the test makers are measuring is your ability to distinguish "true facts" stated in the problem from unsupported assumptions made out of carelessness or inattention. It's likely that several data sufficiency items will test this ability.

Don't assume accuracy of diagrams

The test makers go out of their way to warn you about their diagrams. There are a couple of specific considerations to bear in mind.
First, any diagram provided will agree with the information stated in the question, but not necessarily with the information provided in either of the numbered statements.
Second, the diagram may or may not be drawn to scale. You can safely assume that lines do, in fact, touch where they appear to touch; you can also assume that lines that appear straight are straight.
But you can't make any assumptions about the size of angles, the length of lines, the regularity of figures, etc.

1.4 Data Sufficiency Examples

Ex1. If n is a prime number between 10 and 20, what is the value of n?

A. $n > 12$
B. $n < 17$

Sol: In this question, you are being asked to determine the value of n. This means the exact value that n represents. One and only one number is an acceptable answer to a question so phrased. What would you need to know to answer this? If you know the definition of a prime number, then you can quickly figure out that there are four prime numbers between 10 and 20 ; 11, 13, 17 and 19. Now consider the two statements and see whether either or both can give you the information you need. Firstly statement A, it tells you that n is greater than 12, which does not enable you to answer the question uniquely. It still leaves three possible answers (13, 17 and 19). Hence statement A alone is not sufficient. Now statement B by itself is also not sufficient as if n is less than 17, there are still two possible answers (11 and 13). Finally, if you combine both the statements, you know that n is greater than 12 and that n is less than 17, only one possible number that fits in these conditions is 13. Hence the question can be solved with the help of both statements together.

Ex2. Three packages have a combined weight of 96 kg. What is the weight of the heaviest?

A. One package weighs 24 kg.
B. One package weighs 48 kg.

Sol: Statement A is not sufficient to determine the weight of the heaviest package. It implies only that the combined weight of the other two packages is 72 kg. Statement B alone is sufficient for it implies that the combined weight of two of the packages is only 48 kg. Since the weight of the 48 -kg packages is equal to the combined weight of the other two packages, the heaviest package must weigh 48 kg. So we can see that statement B alone is sufficient to answer the question but statement A alone is not.

Ex3. Is the product of two numbers greater than 100?

A. The sum of the two numbers is greater than 50.
B. Each of the numbers is greater than 10.

Sol: The statement A alone is not sufficient can be proved by examples. If the two numbers are 30 and 31, their sum is greater than 50 and their product is greater than 100; but if the two numbers are 50 and 1, though their sum is greater than 50, their product is only 50, and less than 100. Statement B is sufficient. If both of the numbers are greater than 10, then their product must be greater than 10 x 10, or greater than 100.

Ex4. Is x greater than 0?

A. x^3 is less than 0.
B. $3x = -3$.

Sol: Statement A establishes that $x^3 < 0$, so x itself must be a negative number. Statement A alone, therefore, is sufficient so establish that the answer to the question is “no, x is not greater than 0”. Similarly, statement B alone also establishes that x is negative. Watch out! Some test-takers would, give reasoning (incorrectly) that since the answer to the question is “no”, the information is not sufficient. In fact, the information is sufficient to give a definite negative answer to the question.

Ex5. At a clothing store, Amit spent Rs. 130. How many of the articles of clothing that Amit purchased were priced at Rs. 15?

A. Amit purchased only articles costing Rs. 15 and Rs. 20.
B. Amit purchased more than two Rs. 20 articles.

Sol: It is true that Statement A alone doesn't provide enough information to answer the question, it does narrow the possibilities to two: two Rs. 20 articles plus six Rs. 15 articles, or five Rs. 20 articles and two Rs. 15 articles. And when the information provided in statement B is included, an answer can be obtained. Since the articles of clothing are whole articles (no fraction allowed), Amit purchased exactly two Rs. 15 articles. So the correct answer is obtained by combining the two statements.

Ex6. If x and y are integers, what is the value of x^2?

A. $xy = 7$.
B. $y < 4$.

Sol: It's probably clear that neither statement alone gives enough information to answer the question. However, you might be tempted to answer that both statements combined are sufficient, reasoning as follows: "Statement A tells us that the product of x and y equals 7, and since 7 is a prime number, the only two numbers we can multiply together to get 7 are 1 and 7. Then statement B tells us that y is less than 4.

Combining these two facts, we conclude that y must equal 1, making x equal to 7."
Not so! The possibility overlooked here is that x and y might both be negative integers, specifically, –1 and –7. When multiplied together, these two integers have a product of +7, just as statement A states. And both have a value less than 4, in accordance with statement B. Therefore, even when we combine the two numbered statements, we cannot determine with certainty the value of x, which could be equal to 7, –1 or –7. Thus, the answer cannot be obtained even by combining both the statements.

Ex7. Is $xy > 0$?

A. $xy^2 > 0$
B. $x^2y^3 < 0$

Sol: Any quantity raised to the second power (or other positive even power) generates a positive result. But a quantity raised to the third power (or other positive odd power) may be positive or negative depending on the sign of the original quantity. Thus, statement A alone is not sufficient to establish the sign of the xy, but it does establish that x is positive. Statement B alone is not sufficient to establish the sign of xy, but it does establish that y^3 and therefore y is negative. So both statements taken together establish that $xy < 0$. Hence the answer is obtained by combining both the statements.

Ex8. An item is discounted by 15 % from its usual selling price. What is the usual selling price?

A. The value of the discount is Rs. 450.
B. The discounted price is Rs. 2550.

Sol: Statement A alone is sufficient, since 15 % of the usual selling price is Rs. 450, value of the discount:

0.15 × Usual Price = Rs. 450. Usual Price = $\frac{\text{Rs } 450}{0.15}$ = Rs. 3000

Statement B alone is also sufficient, for the discounted price is equal to the usual selling price minus the discount; and the discount can be expressed as a percentage of the usual selling price.

Discount Price = Usual Price – (0.15 Usual Price) ⇒ Discount = 0.85 Usual Price
⇒ 0.85 Usual Price = Rs. 2550 ⇒ Usual Price = Rs. 3000. So each statement alone is sufficient.

Before attempting the questions on this topic in the exam you must solve some data sufficiency test, so that you are able to apply the tips and tricks you have learnt by going through these examples.

Ex9. How many books are there on a certain shelf?

A. If four books are removed, the number of books remaining on the shelf will be less than 12.
B. If three more books are placed on the shelf, the total number of books on the shelf will be more than 17.

Sol: Neither statement alone is sufficient to answer the question asked. Statement A alone implies only that the number of books on the shelf is 15 or fewer, and statement B alone implies only that the number of books on the shelf is 15 or more. But the two statements taken together are sufficient to answer the question, for they imply that the number of books on the shelf is 15. (15 is the only integer that satisfies both statements A and B). Hence the answer is obtained by combining both the statements.

Ex10. What is the average weight of the five starting players on the defensive line of the Mammoths football team?

A. The three heaviest players on the line have an average weight of 340 pounds.
B. The two lightest players on the line weigh 275 and 290 pounds, respectively.

Sol: To calculate the average of a group of numbers, you need just two pieces of information: the total of the numbers, and the number of numbers. In this case, the only missing piece of information is the total weight of the Mammoths defensive line. (You already know the number of numbers involved - five, since there are five players on the line.)

Neither numbered statement alone gives you the players' total weight, but if you combine statements A and B, you can determine it. You'd multiply 340 by 3 (to get the combined weight of the three heaviest players) and add 275 and 290 (the weights of the two lightest players). But whatever you do, don't actually perform these steps. It's not necessary. All that matters is that you can tell that it would be theoretically possible to make these calculations and so determine the average. Hence the answer is obtained by combining both the statements.

Ex11. What is x?

A. $x + 2 = 4$
B. $x^2 = 4$

Sol: Statement A alone is sufficient to establish the exact value of x as 2. Statement B, however, is not sufficient. If $x^2 = 4$, x can be either + 2 or –2, and that is not sufficient to answer a question that asks "What is x?" The answer is option 1.

Ex11. What is x?

A. $x^2 - 1 = 0$
B. $x = (10)^0$

Sol: Statement B alone is sufficient to peg the exact value of x. Since any number to the zero power equals 1, $10^0 = 1$, and x must be 1. Statement A alone is not sufficient. Statement A establishes that either $x + 1 = 0$ or $x - 1 = 0$, so that $x \pm 1$. But knowing that x is one of two possible values is not sufficient to answer a question that asks for the value of x. Hence the answer is option 2.

Ex12. What is the maximum number of cubic blocks of wood that will fit into a box?

A. The edge of each block is 2 inches long.
B. The box has the shape of a rectangular solid with inner dimensions 20 inches by 40 inches by 16 inches.

Sol: To determine exactly how many blocks the box will hold, we need the size and shape of both the blocks and the box. Neither statement A nor statement B alone will give you all of the information you need. Statement A gives you the size and shape of the blocks while statement B gives you the size and shape of the box. Both taken together give all of the information you need. The answer is option 3.

Ex13. What is the area of Circle O?

A. The diameter of Circle O is 4.
B. The circumference of Circle O is 4π.

Sol: Statement A alone is sufficient, for knowing the diameter of the circle allows you to determine the radius, and in turn, the radius gives you the area of the circle by the formula Area = πr^2. Statement B alone is also sufficient. The circumference is 4π. Since the formula for the circumference is $C = 2\pi r$, $4\pi = 2\pi r$, and $r = 2$. So statement A alone is sufficient and statement B alone is sufficient to give the exact value of the area of the circle. The answer is option 4.

Ex14. The total number of employees on the payroll of Corporation X was what percent greater on June 30 than it had been on June 1 of the same year?

A. During the month of June, 15 employees were dropped from the payroll.
B. During the month of June, 37 employees were added to the payroll.

Sol: The question asks for the percent increase in the number of employees during June. The two statements together establish that 37 - 15 = 22 people were added to the payroll during June, but that is not sufficient to answer the question asked. Hence the answer is option 5.

Ex15. If the average height of three people is 68 inches, is the shortest person more than 60 inches tall?

A. The height of the tallest person is 72 inches.
B. One of the persons is 70 inches tall.

Sol: The answer is option 1. Statement A alone is sufficient to answer the question with a definite yes. If x, y, and z represent the three heights, then $(x + y + z) / 3 = 68$. So $x + y + z = 204$
Then if one of the heights, say z, is 72: $x + y = 132$.
This means that the sum of the heights of the other two persons must be 132; but we know that neither of them can be as tall as 72 inches, so the shortest person must be taller than 132 – 72 = 60 inches. Statement B by itself is not sufficient since it may or may not refer to the tallest person.

Ex16. Is the product of two numbers greater than 100?

A. The sum of the two numbers is greater than 50.
B. Each of the numbers is greater than 10.

Sol: The answer is option 2. That statement A alone is not sufficient can be proved by examples. If the two numbers are 30 and 31, their sum is greater than 50 and their product is greater than 100; but if the two numbers are 50 and 1, though their sum is greater than 50, their product is only 50, and less than 100. Statement B is sufficient. If both of the numbers are greater than 10, then their product must be greater than 10 x 10, or greater than 100.

Ex17. Is x a positive integer?

A. $x > 0$.
B. $x^2 + 16 = 25$.

Sol: The answer is option 3. Statement A alone is not sufficient, for it establishes only that x is positive but has nothing to say as to whether x is an integer. Statement B alone is not sufficient, for it establishes that x is an integer but fails to establish whether x is + 3 or -3. Both taken together establish that $x = +3$, so they answer the question: x is a positive integer. Hence the answer is option 3.

Ex18. Is x greater than 0?

A. x^3 is less than 0.
B. $3x = -3$.

Sol: The answer is option 4. Statement A establishes that $x^3 < 0$, so x itself must be a negative number. Statement A alone, therefore, is sufficient so establish that the answer to the question is "no, x is not greater than 0". Similarly, statement B alone also establishes that x is negative. Watch out! Some test-takers would call this an option 5, reasoning (incorrectly) that since the answer to the question is "no", the information is not sufficient. In fact, the information is sufficient to give a definite negative answer to the question. Thus the answer is option 4.

Ex19. Is a cubic centimeter of substance S heavier than a cubic centimeter of substance T?

A. A cubic centimeter of S weighs more than 0.25 cubic centimeters of T.
B. 3 cubic centimeters of T weighs less than 5 cubic centimeters of S.

Sol: The answer is option 5. Statement A establishes only that a cubic unit of S is heavier than 0.25 cubic units of T, but it does not establish how much heavier. Statement B works in the same way. Even taking them together, it is not possible to establish which substance is heavier. For example, S and T might even have equal weights.

Ex20. At a clothing store, Fred spent $130. How many of the articles of clothing that Fred purchased were priced at $15?

A. Fred purchased only articles costing $15 and $20.
B. Fred purchased more than two $20 articles.

Sol: At first glance, you might think that the answer to this question is option 5. But you should look a little closely. Although it is true that Statement A alone doesn't provide enough information to answer the question, it does narrow the possibilities to two: two $20 articles plus six $15 articles, or five $20 articles and two $15 articles. And when the information provided in statement B is included, an answer can be obtained. Since the articles of clothing are whole articles (no fraction allowed), Fred purchased exactly two $15 articles. So the correct answer is option 3.

Ex21. Is $\frac{x}{15}$ an integer?

A. $\frac{x}{3}$ is an integer.

B. $\frac{x}{5}$ is an integer.

Sol: Neither Statement A nor statement B alone is sufficient to answer the question. As for statement A, even though x is divisible by 3, x might not be divisible by 15. (For example, if x is 9.) Similarly, statement B alone is not sufficient. Even though x is divisible by 5, x might not be divisible by 15. (For example, if x is 10.) But both statements taken together are sufficient to answer the question. If x is divisible by both 3 and 5, then x must be divisible by 15. Hence the correct answer is option 3.

Ex22. What is the one-thousandth term in sequence *S*?

A. The fifth term is *S* is 47.
B. Each term in *S* following the first term is generated by multiplying the preceding term by 4 and adding 1.

Sol: The rule for constructing the sequence is given in statement B. And the point of reference is given in statement A. There is no need to try to find the value of the thousandth term. You need only recognize that it is possible to do so, so the answer must be option 3.

Ex23. If *x* and *y* are positive numbers, what percentage is *x* of *y*?

A. The ratio *x* : *y* is 5:4

B. *y* is $\frac{4}{5}$ of *x*.

Sol: Statement A is sufficient. Since the ratio of *x* to *y* is 5 to 4, $\frac{x}{y} = \frac{5}{4}$ =1.25 –125 %. Statement B is also sufficient. Since $y = \frac{4}{5}x$, $\frac{x}{y} = \frac{5}{4}$ = 1.25 =125 %. So the correct answer is option 4.

Similarly, if you know fractional, percentage, or ratio relationships between two quantities, you can also determine the reciprocals of those relationships.

Thus if you know that x is 125% of y, you can deduce that y is 80% of x: $x = 1.25y$, so $x = \frac{5y}{x}$ and $\frac{y}{x} = \frac{4}{5} = 80\ \%$.

Ex24. Did Diane receive a larger bonus than Claire?

A. Diane's bonus was 7 % of her annual salary.
B. Claire's bonus was 8 % of her annual salary.

Sol: To find the actual size of either bonus, you would need to know the dollar amount for the annual of both salaries. As it is, you know only that Claire's bonus was a larger percentage of her salary than that of Diane's bonus. That is not sufficient to answer the question asked, so the correct answer is option 5.

Ex25. Which of three books, X, Y, or Z costs the least?

A. The cost of X is $\frac{2}{3}$ the cost of Y.

B. The cost of Z is $\frac{5}{4}$ the cost of X.

Sol: Neither statement provides the actual cost of any of the three books, but both statements taken together do establish that X is least expensive, statement A establishes that X is less than Y, and statement B establishes that X is less than Z. So the correct answer is option 3.

Ex26. An item is discounted by 15 % from its usual selling price. What is the usual selling price?

A. The dollar value of the discount is $45.
B. The discounted price is $255.

Sol: Statement A alone is sufficient, since 15 % of the usual selling price is the $ 45 value of the discount:

0.15 × Usual Price =$45

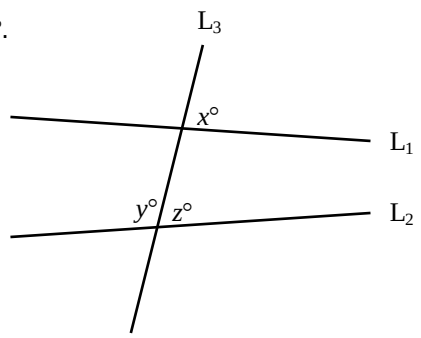
Usual Price = $\frac{\$45}{.15}$ =$300

Statement B alone is also sufficient, for the discounted price is equal to the usual selling price minus the discount; and the discount can be expressed as a percentage of the usual selling price.
Discount Price = Usual Price – (0.15 Usual Price)
Discount = 0.85 Usual Price
0.85 Usual Price = $255
Usual Price = $300

So each statement alone is sufficient, and the correct answer is option 4.

Ex27. In the figure given below, are lines L_1 and L_2 parallel to each other?

A. $x = z$.
B. $x + y = 180°$.



The correct answer is option 4. Although the lines do not appear to be parallel, you should not conclude on the basis of the figure that they are not parallel. In fact, each statement establishes that they are parallel.

Ex28. If $xy = 0$, is x greater than y?

A. $4x > 5y$
B. $x < 0$

Sol: The correct answer is option 4. In statement A, if x is +ve then y must be 0 and if x is 0 then y must be -ve for the product to be 0. So statement A alone is sufficient. In statement B, if x is -ve then y has to be 0 to make the product 0 and then $x < y$. So statement B alone is also sufficient.

Ex29. Is $xy > 0$?

A. $xy^2 > 0$
B. $x^2y^3 < 0$

Sol: The correct answer is option 3. Any quantity raised to the second power (or other positive even power) generates a positive result. But a quantity raised to the third power (or other positive odd power) may be positive or negative depending on the sign of the original quantity. Thus, statement A alone is not sufficient to establish the sign of the xy, but it does establish that x is positive. Statement B alone is not sufficient to establish the sign of xy, but it does establish that y^3 and therefore y is negative. So both statements taken together establish that $xy < 0$.

Ex30. What is value of x?

A. $x^2 - 9 = 0$
B. $x > 0$

Sol: The correct answer is option 3. Statement A alone is not sufficient to fix the value of x. Since $x^2 - 9 = 0$, $x^2 = 9$, and $x = +3$ or $x = -3$. That conclusion coupled with statement B, however, is sufficient to answer the question.

Ex31. What is the value of x?

A. $x + y = 12$.
B. $2y = 6$.

Sol: The correct answer is option 3. Neither statement alone is sufficient to fix the value of x. But if you solve for y in the second equation ($y = 3$) and substitute that value for y into the first equation ($x + 3 = 12$), you can find the value of x.

Ex32. If $x + y \neq 0$, what is the value of $\frac{x^2 - y^2}{x + y}$?

A. $x - y = 2$.
B. $x + y = 2$.

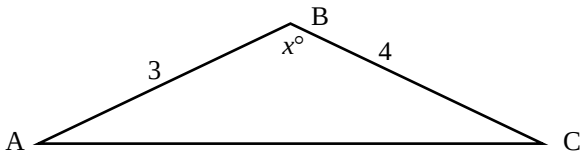
Sol: The correction answer is option 1. Factor the numerator of the expression and simplify:

$$\frac{x^2 - y^2}{x + y} = \frac{(x + y)(x - y)}{x + y} = x - y$$

So statement A alone is sufficient to answer the question, but statement B alone is not. So the correct answer is option 1.

Ex33. What is the value of x in the figure given below?

A. $AC^2 = AB^2 + BC^2$.
B. AC = 5.



Sol: The Pythagorean theorem is a favourite:
The correct answer is option 4. Statement A, which is the Pythagorean formula, establishes that the sides of the triangle create a right triangle and that AC is the hypotenuse. So $x = 90°$. Statement B also establishes that the triangle is a right triangle. (It has sides of 3, 4 and 5). So $x = 90°$.

Ex34. If x and y are integers, what is the value of x^2?

A. $xy = 7$.
B. $y < 4$.

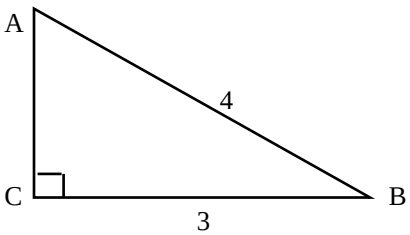
Sol: It's probably clear that neither statement alone gives enough information to answer the question. However, you might be tempted to choose option 3, reasoning as follows: "Statement A tells us that the product of x and y equals 7, and since 7 is a prime number, the only two numbers we can multiply together to get 7 are 1 and 7. Then statement B tells us that y is less than 4.
Combining these two facts, we conclude that y must equal 1, making x equal to 7."
Not so! The possibility overlooked here is that x and y might both be negative integers, specifically,– 1 and –7. When multiplied together, these two integers have a product of +7, just as statement A states. And both have a value less than 4, in accordance with statement B. Therefore, even when we combine the two numbered statements, we cannot determine with certainty the value of x, which could equal either 1, –1 or –7. Thus, the correct answer is option 5.

Ex35. What percentage of the student in a certain history class is majoring in economics?

A. 64 % of the male students in the class are majoring in economics.
B. 42 % of the female students in the class are majoring in economics.

Sol: The question would be solvable, and easily so, if the number of male and female students in the class was equal (or, for that matter, if we knew the proportion of male to female students). However, we can't assume that the number of males and the number of females is equal, so the answer is option 5.

Ex36. In triangle ABC, what is the length of side AC?

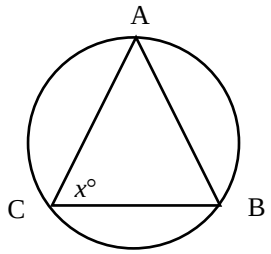


A. Triangle ABC is a right triangle.
B. The length of AB is 4 and the length of BC is 3.

Sol: Eager to use your knowledge of the Pythagorean theorem and the famous "Pythagorean triple" triangles, you may assume that this is a 3-4-5 right triangle, with side AC being the hypotenuse. However, we can't actually tell that from the information given. It's possible that the triangle might look something like this:
In this case, with side AC the shorter leg rather than the hypotenuse the Pythagorean theorem tells us that side AC would have a length of $\sqrt{7}$, equal to about 2.65 units. Since there is more than one possible solution, this question, too, should be answered as option 5.

Ex.37. Is the triangle in the figure given below equilateral?

A. Arc AB has a degree measure of 120°.
B. $x = 60°$.



Sol: The triangle inscribed in the circle certainly looks equilateral in the diagram provided. The real or apparent dimensions of the triangle as shown in the diagram are irrelevant to answering the question. Only the facts given in the question and the two numbered statements can be taken seriously.

In this case, we have another answer as option 5 – the two statements taken together are not sufficient. Both statement A and statement B tell us the same thing: that $\angle C$ has a measure of 60° – as it would if the triangle were equilateral. But without further information, we can't tell that angles A and B are also 60° angles. It's possible that angle A measures 59° while angle B measures 61°, in which case this is not an equilateral triangle. Since we can't know for sure the answer must be option 5.

Data Caselets

2.1 Data Caselets

Caselets are basically a set of linked questions -- there are two to five questions based on the same information and the questions are related in some manner. Very often, caselets provide data within a paragraph or a set of statements, rather than in the form of a table or graph. Caselets vary in length, as the amount of information contained in provided in different sentences and paragraphs. While reading a caselet, it is always advisable to underline the important facts and figures and if necessary, one can make his/her own table/chart/graph for solving the questions.

Preparing the table can be time consuming and effort intensive. Hence, it is strongly advised that students regularly practice a lot for these types of questions.

Hence the solution to the caselet often lies in making the correct table/chart for the data. There will be missing numbers in the table, and so you will have to consider various possibilities for the possible values of these missing numbers. Caselet questions are not easy to manage at times as the interpretation of required data in a meaningful manner is left to student's ability to comprehend the hidden facts. In CAT and other competitive exams, caselets can be asked either/both in Quant and Data Interpretation section and it is similar to the Reading Comprehension part in English Usage section. The best way to master caselets is to practice and practice. The below given examples would help you to understand the kind of questions, which appear in exams and approach you should apply to solve those questions.

Illustration 1: Mohan took voluntary retirement in Dec. 2011 and received a certain amount of money as retirement benefits. On Jan 1, 2012, he invested the entire amount in shares. At the end of the month, he sold all his shares and realized 25% profit. On Feb 1, he reinvested the entire amount in shares which he sold at the end of the month at a loss of 20%. Again, he invested the entire amount on Mar 1 in a new company. At the end of the month, he sold the shares of the new company to a friend and realised a profit of 20% in the process. He invested the entire amount in shares on Apr 1, which he sold at the end of the month for Rs. 1,08,000 incurring a loss of 10%.

1. What is the amount of retirement benefits received by Mohan?

1. Rs.1,08,000 2. Rs.1,25,000 3. Rs. 1,20,000 4.Rs. 1,00,000

Sol: Now as in this data caselet the information starts from the amount he received at the time of his retirement, the question should start from there. So, let the amount received by Mohan in Dec. 2011 be Rs. 100, as retirement benefits:
Therefore, investment in the month of Jan 2012 = 100.
Profit of 25% at the end of Jan 2012.
Hence, investment in the month of Feb 2012 = 125.
Loss of 20% at the end of Feb 2012.
Hence, investment in the month of March 2012 = 100.
Profit of 20% at the end of March 2012.
Hence, investment in the month of April 2012 = 120.
Loss of 10% at the end of April 2012
Therefore the amount left at the end of April 2012 = 108. Now Rs. 100 received initially becomes Rs. 108 at the end of April. What sum of money will amount to Rs. 1,08,000 $\Rightarrow$ 108000 × 100/108 = 1,00,000.

2. The percentage profit received by Mohan between Jan 1 and Apr 30 is:

1. 8.00% 2. 15.00% 3. -10.00% 4. None of these

Sol: As we calculated above, that Rs. 100 received would have become Rs. 108 i.e. he gets a profit of Rs. 8 on his initial investment of Rs. 100. The percentage profit earned by him will be 100 × 8/100 = 8%.

3. The amount of loss incurred by Mohan based on his operation in Apr 2012 is:

1. Rs. 25,000 2. Rs. 12,000 3. Rs. 20,000 4. Rs. 8,000

Sol: As solved in the first question, he invested Rs. 120 in the month of April. Now as per the solution of this equation, Rs. 120 would be Rs. 120000 in total as 100 corresponds to Rs. 100000. When he invested Rs. 120000, he gets a loss of 10% in the month of April. So the loss made by him is 120000 × 10/100 = Rs. 12000.

4. The maximum amount invested by Mohan in any one month was in:

1. January 2. February 3. March 4. April

Sol: Now as in this data caselet the information starts from the amount he received at the time of his retirement, the question should start from there. So, let the amount received by Mohan in Dec. 2011 be Rs. 100, as retirement benefits:

Therefore, investment in the month of Jan 2012 = 100.
Profit of 25% at the end of Jan 2012.
Hence, investment in the month of Feb 2012 = 125.
Loss of 20% at the end of Feb 2012.
Hence, investment in the month of March 2012 = 100.
Profit of 20% at the end of March 2012.
Hence, investment in the month of April 2012 = 120.
Loss of 10% at the end of April 2012.

Now as per this hypothetical value, he invested the largest amount in the month of February, which was Rs. 125. Let us calculate the corresponding value. If Rs. 108 in this assumed value is actually Rs. 108000, so Rs. 125 will be ⇒ 125 × 108000/108 = Rs. 125000.

Illustration 2: An institute offers MBA with specialization in Marketing, Finance and HR. Among the total number of students in the institute, 45% are girls. Number of boys studying Marketing is 30% of the total number of boys in the institute which is 297. 40% of the girls are studying HR. Number of boys and girls studying Marketing are in the ratio of 3: 2. 50% of boys are studying Finance.

1. How many girls are studying Finance?

Sol: Number of boys studying Marketing = 297. It is given that 30% of the boys in the institute study Marketing. Let us calculate the total number of boys in the institute = (297 × 100)/30 = 990. It is given in the question, that there are 45% girls in the institute, which implies there are 55% boys. Now if the total number of boys is 990, the total number of girls will be (45/55) × 990 = 810. Number of girls studying HR are 40% of the total number of girls, which becomes = 810 × 40/100 = 324. The ratio of number of boys and number of girls studying Marketing is given to be 3 : 2. The number of boys is given to be 297. So the number of girls studying Marketing would be = 297 × 2/3 = 198. Number of boys studying Finance are half of the total (i.e. 50%, given in the question) = (1/2) × 990 = 495. Number of boys studying HR can be found by subtracting the number of boys studying Marketing and Finance from the total number of boys and there are three kinds of specialization are available only i.e. = 990 – 297 – 495 = 198. Number of girls studying Finance can also be found in the same manner = 810 – 324 – 198 = 288.

2. Number of girls studying Marketing is what percent of the boys studying Finance?

Sol: As calculated above the number of girls studying Marketing is 198 and the number of boys studying Finance is 495. The required percentage is = (198/495) × 100 = 40%.

3. Number of boys studying Finance is what percent of the total number of the students in the Institute?

Sol: Total number of the students in the Institute = 990 + 810 = 1800.
The number of boys studying Finance = 495. Now the percentage will be 100× 495/1800 = 27.5%.

4. What is the respective ratio between number of boys and girls studying HR?

Sol: The number of boys studying HR is 198 and the number of girls studying HR is 324.
Required ratio = 198 : 324 = 11 : 18.

2.2 Data Caselets Examples

DIRECTIONS ***for questions 1 to 4:*** *Answer the questions on the basis of the information given below.*

The employees of Preliance group of companies are undergoing an intercompany promotional exam. In all 198 employees from Communications(C), Infrastructure (I), Telecom (T) and Petrochemicals (P) are eligible for the exam. The breakup of number of employees in these companies is in such a way that the difference between the number of employees in any two companies is at most 30. The ratio of the number of employees in Communications and Infrastructure is 5 : 6, which is same as the ratio of number of employees in Telecom and Petrochemicals. All the employees wrote a promotional exam in which the maximum marks are 100. The average marks scored by the employees in each company are calculated.

It is known that if the companies were ranked according to the average marks of the employees in the company, these would be ranked 2, 1, 3 and 4 respectively, with rank 1 allotted to the company with the highest average marks, the next one second and so on. In the same manner if the companies are ranked according to the total marks of the employees in the company, they would be ranked 4, 2, 3 and 1 respectively.

It is also known that the number of employees in Communications is less than that of Telecom.
The marks scored by each employee and the average marks of each company are integers.
The average marks of each company are at least 50.

1. If the average marks of Communications are 74, then the average marks of Petrochemicals are at least

1. 60 2. 61 3. 64 4. 62

Sol: Let the number of employees in C, I, T and P companies be 5x, 6x, 5y, and 6y respectively. $\Rightarrow$ 5x + 6x+ 5y + 6y = 198 $\Rightarrow$ x + y = 18 $\rightarrow$ (1) and 6y – 5x $\leq$ 30 $\rightarrow$(2)
From (1) and (2): x = 8 and y = 10. Number of employees in the companies:
C = 40, I = 48, T = 50, P = 60
Average marks of company C = 74
The least average marks of company I = 75
The least total marks of company I = 48 × 75 = 3600
But the total marks of company P > 3600
Average marks of company P > 3600/60 i.e., > 60
$\therefore$ the average marks of company P are at least 61.

2. If the difference between the total marks scored by employees in Infrastructure and Telecom are 100, then what can be the difference between the average marks, of these companies?

1. 8 2. 6 3. 7 4. 9

Sol: Let the number of employees in C, I, T and P companies be 5x, 6x, 5y, and 6y respectively. $\Rightarrow$ 5x + 6x+ 5y + 6y = 198 $\Rightarrow$ x + y = 18 $\rightarrow$ (1) and 6y – 5x $\leq$ 30 $\rightarrow$(2)
From (1) and (2): x = 8 and y = 10. Number of employees in the companies:
C = 40, I = 48, T = 50, P = 60
Let the average marks of the companies I and T be i and t respectively.
$\Rightarrow$ 48i – 50t = 100 $\rightarrow$ (1) and i $\geq$ 50; t $\geq$ 50 $\rightarrow$ (2)
From (1) and (2) there are two possibilities.
(i) i = 75 and t = 70 and i – t = 5.
(ii) i = 100 and t = 94 and i – t = 6.
$\Rightarrow$ Now in the options, only 6 is given, which will be our answer.

3. If five persons from Telecom whose average marks are more than the initial average of Telecom are shifted to Communications, then how many ranking orders are possible if the companies are ranked by average marks?
(Assume that no two companies will have the same average after the arrangement)

1. 2 2. 4 3. 8 4. 12

Sol: Let the number of employees in C, I, T and P companies be 5x, 6x, 5y, and 6y respectively. $\Rightarrow$ 5x + 6x+ 5y + 6y = 198 $\Rightarrow$ x + y = 18 $\rightarrow$ (1) and 6y – 5x $\leq$ 30 $\rightarrow$(2)
From (1) and (2): x = 8 and y = 10. Number of employees in the companies:
C = 40, I = 48, T = 50, P = 60
According to the given information, the average marks of T may become less than that of P. And the average marks of C may become more than I.
$\therefore$ Four different arrangements P_1, P_2, P_3 and P_4 are possible

	P_1	P_2	P_3	P_4
C	2	1	2	1
I	1	2	1	2
T	4	3	3	4
P	3	4	4	3

4. Which of the following could be the ratio of the averages of the marks scored by the employees of Communications, Infrastructure, Telecom and Petrochemicals respectively?

1. 18:19:16:17 2. 18: 19:16:15 3. 18:19:17:16 4. 18:20:17:16

Sol: Let the number of employees in C, I, T and P companies be 5x, 6x, 5y, and 6y respectively. $\Rightarrow$ 5x + 6x+ 5y + 6y = 198 $\Rightarrow$ x + y = 18 $\rightarrow$ (1) and 6y – 5x $\leq$ 30 $\rightarrow$(2)
From (1) and (2): x = 8 and y = 10. Number of employees in the companies:
C = 40, I = 48, T = 50, P = 60
From the given options, option (3) satisfies both the given ranking orders

DIRECTIONS *for questions 5 to 9:* *Answer the questions on the basis of the information given below.*

The following are the marks of five students - A, B, C, D and E taken three at a time - 250, 254, 256, 264, 266, 270, 270, 272, 276 and 286. The marks of each student is a distinct integer. It is known that E has the maximum marks and A, B and D together have 32 marks less than C, D and E together.

5. What is the average marks of the five students?

1. 88.8 2. 88 3. 90.4 4. 92.6

Sol: It is given that the difference between the marks of A, B, D and C, D, E is 32. From the given values we know A + B + D = 254 and C + D + E = 286 is the only option. As 254 is the second lowest among all the marks given, it is obtained when the two smallest and the second largest marks are added. As 286 is the maximum among all the marks so it is obtained when the three highest marks are added. As D is common to both the groups, D has the second highest marks and C has the third highest marks. A and B would have the least and second least marks in any order.
Adding all the marks we get 6 (A + B + C + D + E) = 250 + 254 + 256 + 264 + 266 + 270 + 270 + 272 + 276 + 286 = 2664 $\therefore$ A + B + C + D + E = 444
For ease of calculation, let us assume that B has more marks than A.
A + B + C = 250 $\therefore$ D + E = 194 C + D + E = 286 $\therefore$ C = 92
As we have assumed B > A, so the second highest group has to be B + D + E = 276
$\therefore$ B = 82 A = 250 – (82 + 92) = 76
The second lowest group has to be A + B + D
A + B + D = 254 D = 254 – (76 + 82) = 96.
$\therefore$ E = 194 – 96 = 98
$\therefore$ The marks are A – 76, B – 82, C – 92, D – 96 and E – 98. (A and B could also be 82 and 76 respectively)
The total marks of all the students = 76 + 82 + 92 + 96 + 98 = 444. Hence the average marks = 444/5 = 88.8

6. Which student has third highest marks?

1. A 2. D 3. C 4. Can't be determined

Sol: It is given that the difference between the marks of A, B, D and C, D, E is 32. From the given values we know A + B + D = 254 and C + D + E = 286 is the only option. As 254 is the second lowest among all the marks given, it is obtained when the two smallest and the second largest marks are added. As 286 is the maximum among all the marks so it is obtained when the three highest marks are added. As D is common to both the groups, D has the second highest marks and C has the third highest marks. A and B would have the least and second least marks in any order.

Adding all the marks we get 6 (A + B + C + D + E) = 250 + 254 + 256 + 264 + 266 + 270 + 270 + 272 + 276 + 286 = 2664 ∴ A + B + C + D + E = 444

For ease of calculation, let us assume that B has more marks than A.

A + B + C = 250 ∴ D + E = 194 C + D + E = 286 ∴ C = 92

As we have assumed B > A, so the second highest group has to be B + D + E = 276

∴ B = 82 A = 250 – (82 + 92) = 76

The second lowest group has to be A + B + D

A + B + D = 254 D = 254 – (76 + 82) = 96.

∴ E = 194 – 96 = 98

∴ The marks are A – 76, B – 82, C – 92, D – 96 and E – 98. (A and B could also be 82 and 76 respectively)

C has the third highest marks.

7. What is the maximum marks of a student?

1. 100 2. 98 3. 94 4. 96

Sol: It is given that the difference between the marks of A, B, D and C, D, E is 32. From the given values we know A + B + D = 254 and C + D + E = 286 is the only option. As 254 is the second lowest among all the marks given, it is obtained when the two smallest and the second largest marks are added. As 286 is the maximum among all the marks so it is obtained when the three highest marks are added. As D is common to both the groups, D has the second highest marks and C has the third highest marks. A and B would have the least and second least marks in any order.

Adding all the marks we get 6 (A + B + C + D + E) = 250 + 254 + 256 + 264 + 266 + 270 + 270 + 272 + 276 + 286 = 2664 ∴ A + B + C + D + E = 444

For ease of calculation, let us assume that B has more marks than A.

A + B + C = 250 ∴ D + E = 194 C + D + E = 286 ∴ C = 92

As we have assumed B > A, so the second highest group has to be B + D + E = 276

∴ B = 82 A = 250 – (82 + 92) = 76

The second lowest group has to be A + B + D

A + B + D = 254 D = 254 – (76 + 82) = 96.

∴ E = 194 – 96 = 98

∴ The marks are A – 76, B – 82, C – 92, D – 96 and E – 98. (A and B could also be 82 and 76 respectively)

The maximum marks are 98.

8. What is the difference in the marks of B and D?

1. 20 2. 10 3. 14 4. Can't be determined

Sol: It is given that the difference between the marks of A, B, D and C, D, E is 32. From the given values we know A + B + D = 254 and C + D + E = 286 is the only option. As 254 is the second lowest among all the marks given, it is obtained when the two smallest and the second largest marks are added. As 286 is the maximum among all the marks so it is obtained when the three highest marks are added. As D is common to both the groups, D has the second highest marks and C has the third highest marks. A and B would have the least and second least marks in any order.

Adding all the marks we get 6 (A + B + C + D + E) = 250 + 254 + 256 + 264 + 266 + 270 + 270 + 272 + 276 + 286 = 2664 ∴ A + B + C + D + E = 444

For ease of calculation, let us assume that B has more marks than A.

A + B + C = 250 ∴ D + E = 194 C + D + E = 286 ∴ C = 92

As we have assumed B > A, so the second highest group has to be B + D + E = 276

∴ B = 82 A = 250 – (82 + 92) = 76

The second lowest group has to be A + B + D
A + B + D = 254 D = 254 – (76 + 82) = 96.
∴ E = 194 – 96 = 98
∴ The marks are A – 76, B – 82, C – 92, D – 96 and E – 98. (A and B could also be 82 and 76 respectively)
The difference in the marks of B and D could be 14 or 20.

9. The maximum difference between the marks of any two of the given students is

1. 18 2. 20 3. 22 4. 16

Sol: It is given that the difference between the marks of A, B, D and C, D, E is 32. From the given values we know A + B + D = 254 and C + D + E = 286 is the only option. As 254 is the second lowest among all the marks given, it is obtained when the two smallest and the second largest marks are added. As 286 is the maximum among all the marks so it is obtained when the three highest marks are added. As D is common to both the groups, D has the second highest marks and C has the third highest marks. A and B would have the least and second least marks in any order.
Adding all the marks we get 6 (A + B + C + D + E) = 250 + 254 + 256 + 264 + 266 + 270 + 270 + 272 + 276 + 286 = 2664 ∴ A + B + C + D + E = 444
For ease of calculation, let us assume that B has more marks than A.
A + B + C = 250 ∴ D + E = 194 C + D + E = 286 ∴ C = 92
As we have assumed B > A, so the second highest group has to be B + D + E = 276
∴ B = 82 A = 250 – (82 + 92) = 76
The second lowest group has to be A + B + D
A + B + D = 254 D = 254 – (76 + 82) = 96.
∴ E = 194 – 96 = 98
∴ The marks are A – 76, B – 82, C – 92, D – 96 and E – 98. (A and B could also be 82 and 76 respectively)
The difference between the marks is maximum for A (or B) and E = 98 – 76 = 22.

Data Comparison

Data Comparison questions test your knowledge of basic math facts and skills along with reasoning, analytical and problem solving abilities not unlike those you may be called to use in business. Each Data Comparison question, instead of solving for a particular value, you need to compare two quantities. At the on set, the Data Comparison question way appears really difficult because of their unfamiliar format. However, once one gets used to them, they can be quicker and easier to solve than the other type of math questions.

The difficulty of the Data Comparison will depend on how well you are doing in the section. In each question, you see two mathematical expressions. They are boxed one in Column A, the other in Column B. Your job is to compare them. Some questions include additional information about one or both quantities. This information is centered, unboxed and essential to making the comparison.

The Directions: Each Data Comparison question consists of two quantities, one in column A and another is column B.
You have to decide, the data given is which column is higher using your knowledge of mathematics and data given, and answer whether

A) If the quantity is column A is higher
B) If the quantity is column B is higher
C) If the two quantities are equal
D) If the relationship cannot be established.

Points to Remember

Choice (A), (B) and (C) all represent definite relationship between the quantities is column A and column B. But choice (D) represents a relationship that cannot be determined. Here are two points to remember about choice (D) that will help when to pick it.

1. Choice (D) is never correct if both columns contain only numbers.
2. Choice (D) is correct if you can demonstrate two different relationships between the columns.

Suppose you ran across the following Data Comparison

Column A	Column B
x	x^2

If x is a number lying between 0 and 1, column A is greater than column B. If x is greater than 1, column B is greater than column A. Because more than one relationship is possible, the answer is (D). As a matter of fact once we establish a second possibility, stop work & pick choice (D)

3.1 Data Comparison Strategy

1. Consider Each Column Individually

This works on Data Comparison that compares two sums or two products.

Illustrative Example:

Column A		Column B
	$w > x > 0 > y > z$	
$w + y$		$x + z$

In this problem we have four variables i.e. w, x, y and z. Compare the value of each 'piece' in the two columns. If every 'piece' is one column is greater than the corresponding 'piece' in the other column then the addition of the columns with greater individual values will have the greater total value.

In this example we know $w > x$ and $y > z$. Therefore, the first term is column A is greater than the first tern is column B. Similarly the second term is the column A is greater than the second term in column B. Conclusively column (A) is greater than the column (B) & the answer is (A).

2. Simplify one to look like the other

This approach helps is cases when columns look so different that you can't compare than directly for example if one column is a fraction and other is a percent converting fraction to percent or vice versa helps.

Illustrative Example:

Column A	Column B
$x(x+2)$	$x^2 + 2x$

In this case column A has brackets and column B doesn't. We can remove brackets of column A
Illustrative Example:

Column A	Column B

The radius of the circle is r and its circumference is R

$2\pi r$	R

Make column B look like column A by rewriting R, the circumference is terms of radius. The circumference of any circle of radius r equals $2\pi r$

3. Treat the two columns like two sides of an inequality

Change both columns by adding, subtracting, multiplying, or dividing by the same amount on both sides to make comparison more apparent. Treat the two columns like two sides of an inequality, with the sign temporarily hidden

Illustrative Example:

Column A	Column B
$4a + 3 = 7b$	
$20a + 10$	$35b - 10$

In this case all the terms of two columns are multiple of 5, so divide by 5 to simplify, we are left with $4a + 2$ is column A and $7b - 1$ in column B. If we add 1 to both the columns, we get $4a + 3$ in column A and $7b$ in column B.
The centered equation tells us thus that the answer is choice (C)

Illustrative Example:

	Column A	Column B
	$y > 0$	
	$1+\frac{y}{1+y}$	$1+\frac{1}{1+y}$
Subtract 1 from both sides	$\frac{y}{1+y}$	$\frac{y}{1+y}$
Multiply both sides by $1 + y$	y	1

Now column (A) is y, which is positive. It also could be a fraction, so one can't get any relation, the answer is (D).

4. Assign Arbitrary Values

Use this to get a handle on abstract algebra data comparison. When the Data Comparison involves variables, assigning arbitrary values helps to make the relationship clearer.

- Assign values that are easy to work with
- Substitute the values and calculate the results. Note the relationship
- Assign different values for each variable and calculate the results.

Column A	Column B
$r > s > k > w > 0$	
$\frac{r}{t}$	$\frac{s}{w}$
$\frac{r}{t} = \frac{4}{2} = 2$	$\frac{s}{w} = \frac{3}{1} = 3$

Let $r = 4$, $s = 3$, $k = 2$ and $w = 1$
So column B is greater than column (A)

Note 1:- Assign more than one value and calculate again. In the above example, we found column (B) was bigger. This does not mean Answer is (B), it only means that answer is not (A) or (C), the answer could be (D)

Let $r = 30$, $s = 3$, $t = 2$, $w = 1$

$\frac{r}{t} = \frac{30}{2} = 15$ $\qquad$ $\frac{s}{w} = \frac{3}{1} = 3$

So column (A) is greater than column (B).
Hence answer is (D).

Note 2:- <u>Assign different kinds of values</u>
Don't assume all variables represent positive integers, unless otherwise specifically mentioned, variables can be zero, negative numbers or fractions. Since different kinds of numbers exhibit different results, always assign different kinds when doing the second round.

3.2 Solved Examples

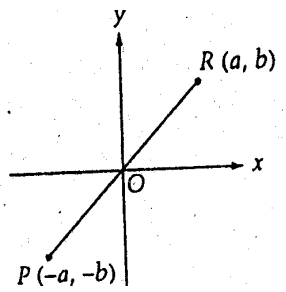
1.

Column A	Column B
$x > y,\ y = z$	
x	y

1. The quantity in Column A is greater
2. The quantity in Column B is greater
3. The two quantities are equal.
4. The relationship cannot be determined.

Sol: Given $x > y$. So answer is (A)

2.



Column A	Column B
The length of segment OP	The length of segment OR

1. The quantity in Column A is greater
2. The quantity in Column B is greater
3. The two quantities are equal.
4. The relationship cannot be determined.

Sol: $= [(a-0)^2 + (b-0)^2]^{\frac{1}{2}} = (a^2 + b^2)^{\frac{1}{2}}$

$OP = [(-a-0)^2 + (-b-0)^2]^{\frac{1}{2}}$

$= (a^2 + b^2)^{\frac{1}{2}}$. So answer is (C).

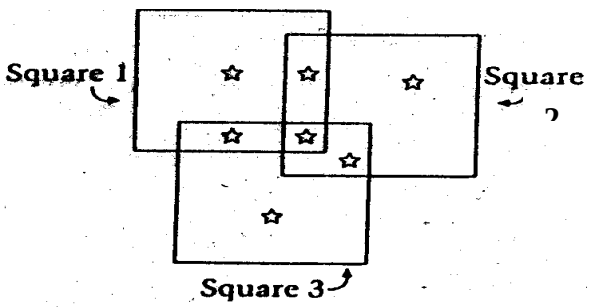
3.

Column A	Column B
One package of muffin mix makes 8 large muffins or 12 small muffins.	
The maximum number of small muffins that can be made with 3 packages of mix	The maximum number of large muffins that can be made with 4 packages of mix

1. The quantity in Column A is greater
2. The quantity in Column B is greater
3. The two quantities are equal.
4. The relationship cannot be determined.

Sol: Small muffins:
1 pack makes 12, 3 packs will make $12 \times 3 = 36$;
Large muffin;
1 pack makes 8, 4 pack will make $4 \times 8 = 32$.
So answer is (A)

4.



Column A	Column B
The number of stars that are in Square 1	The number of stars that are in Square 2 or Square 3 but are not in square 1

1. The quantity in Column A is greater
2. The quantity in Column B is greater
3. The two quantities are equal.
4. The relationship cannot be determined.

Sol: No of stars in square 1 = 4.
Number of stars in square 2 OR square 3 but not in square 1 = 3.
So answer is (A)

5.

Column A	Column B
The circumference of a circle with radius 2	The sum of the circumferences of two circles, each with radius 1

1. The quantity in Column A is greater
2. The quantity in Column B is greater
3. The two quantities are equal.
4. The relationship cannot be determined.

Sol: Circumference of circle = $2 \times 3.14 \times r$,
So for $r = 2$, $C = 2 \times 3.14 \times 2 = (4 \times 3.14)$.
For $r = 1$, $C = 2 \times 3.14 \times 1$.
For two circle of this type $C = (2 \times 3.14 + 2 \times 3.14)$
$= 4 \times 3.14$.
So answer is (C)